

BHAVYASRI K

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate (CGPA: 9.85, Rank 1) with hands-on experience in AI/ML, Agentic AI, Large Language Models (LLMs), Computer Vision, and Full-Stack Development across four internships and award-winning projects. Built production-ready AI systems using Python, TensorFlow, PyTorch, OpenCV, SQL, REST APIs, and Git, including multi-agent research automation, AI-powered digital twins, and computer vision solutions. International award winner (VISAI 2025 – 1st Prize, published patent holder, and AWS Certified AI Practitioner with a passion for building scalable AI products.

TECHNICAL SKILLS

Programming Languages: Python, C, Java, SQL, R

Generative AI & LLMs: LangGraph, LangChain, Ollama, Large Language Models (LLMs), Agentic AI

ML & AI: Supervised/Unsupervised Learning, CNN, LSTM, Time-Series Analysis, TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv8

Web & Software: HTML, CSS, JavaScript, MERN Stack, REST APIs

Databases: MySQL, MariaDB, MongoDB, Firebase, SQLite

Data & Analytics Tools: Pandas, NumPy, Matplotlib, Seaborn, Excel

Tools, Platforms & OS: Linux, Windows, Git, GitHub, PuTTY, WinSCP, Jupyter Notebook, Google Colab

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

INTERNSHIP EXPERIENCE

Mahindra & Mahindra Ltd., Chennai (AI Intern)

Jul– Sep

- Developing Agentic AI solutions for intelligent test automation in automotive software systems.
- Building Python-based AI automation workflows for task planning, execution, validation, and recovery.
- Exploring LLM-based orchestration, AI-driven perception, and context-aware decision-making to improve software testing efficiency.

BorgWarner, Chennai (Deep Learning Intern)

May – Jun 2025

- Developed a YOLOv8-based computer vision system for automated rivet defect detection using custom manufacturing images and webcam-based inference.
- Built and evaluated annotated datasets with OpenCV preprocessing and augmentation to improve model robustness.
- Achieved 0.897 mAP@0.5, 0.758 precision, and 0.891 recall, implementing an end-to-end workflow covering model inference, post-processing, and result visualization.

Edge Matrix Corporation (Remote) (AI/ML Intern)

Dec 2024 – Dec 2025

- Developed a CNN-based plant-disease detection model using 10,000+ labeled images, achieving over 94% accuracy.
- Built a license-plate recognition pipeline using OpenCV and deep learning for real-time detection and character extraction.
- Performed end-to-end ML workflows including preprocessing, feature engineering, hyperparameter tuning, and evaluation using precision, recall, and F1-score.

PROJECTS

Autonomous HMI Validation Using Agentic AI & RAG

- Developed an Agentic AI-based automotive HMI validation system using LLMs, OpenCV, OCR, context-selective RAG, and Python/ADB automation for action planning, validation, recovery, and bug triage.

Low-Cost Digital Twin for Machines (IoT + AI + ML)

- Developed an IoT-based digital twin using ESP32, Raspberry Pi, predictive analytics, and an Ollama-powered agentic AI assistant for real-time machine monitoring, anomaly detection, and intelligent maintenance recommendations.

EV Health Monitoring System (IoT + ML + AR)

- Developed an IoT-enabled EV health monitoring system using Raspberry Pi, predictive analytics, and AR visualization to provide real-time battery diagnostics, fault prediction, and maintenance insights.

Travel Blog Web Application (MERN Stack)

- Built a full-stack MERN travel blog application with secure authentication, REST APIs, dynamic content management, and responsive user interfaces for seamless content publishing and browsing.

EDUCATION

Francis Xavier Engineering College | B.E. Computer Science and Engineering | CGPA: 9.85 | 2023 – 2027

Francis Xavier School (CBSE) | Class XII: 96% (School 1st) | Class X: 95.4% (School 2nd)

Student Mobility Program: Participated in an international academic mobility program at MAHSA University, Malaysia (April 2026).

ACHIEVEMENTS & AWARDS

- VISAI 2025 (International) – 1st Prize (EV Health Monitoring, 200+ teams, Problem Statement by **Mahindra & Mahindra Ltd**)
- VISAI 2026 (International) – 3rd Prize (Digital Twin System, Problem Statement by **Hexagon Manufacturing Intelligence**)
- International Project Expo, Kuala Lumpur – Silver Medal
- Kalasalingam Hack Odyssey 2024 (National) – 3rd Prize

CERTIFICATIONS

- AWS Certified AI Practitioner(AIF -C01)-3 years valid | Deloitte (Forage) – Data Analytics Job Simulation
- NPTEL – Introduction to Internet of Things (82%) | Privacy & Security in Online Social Media (87%)-(Top 1%) | Human Computer Interaction (99%)- (Top 2 %)

PATENT & PUBLICATION

Patent (Published, 2025): IoT-Based Predictive Maintenance System for Intelligent Industrial Automation | Application No.: 202541030745